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Advanced Polymers and Nanostructured Materials Research Letter

Influence of hydrogen peroxide on isotactic polypropylene degradation into micro/nano-plastics under thermal oxidative reaction

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Abstract

Environmental plastic pollution remains a major global concern due to the slow oxidative degradation of micro- and nanoplastics (MPs/NPs). This study explores rapid thermal–oxidative degradation of isotactic polypropylene (iPP) using hydrogen peroxide (H_2O_2) under high-pressure hydrothermal conditions to generate reactive oxygen species (ROS). Complete mineralization occurred at ≥ 27% H_2O_2, while 3–21% induced controlled fragmentation into MPs/NPs with tunable surface oxidation. SEM, FT-IR, XPS, DSC, and Py-GC/MS analyses confirmed chain scission, formation of carbonyl and hydroxyl groups, reduced crystallinity, and suppression of propylene trimer formation. Pressurized hydrothermal conditions enhanced oxidation kinetics, shortening degradation from years to hours. Overall, high-pressure H_2O_2 oxidation effectively accelerates polymer breakdown and limits secondary pollutants, presenting a promising route for sustainable MPs/NPs removal in aquatic systems.

Introduction

Today, plastics are widely used everywhere in the world due to the advantage of good plasticity, durability, corrosion resistance, and low cost. From the 1950 s, global plastic production exceeds 400 million tons per year and is expected to reach a cumulative total of 25 billion tons at the end of 2050. However, of the remaining approximately 400 million tons of plastic waste, only 9% are recycled, and another 79% are dumped in landfills and 12% are discarded.[1] Because of improper management, a large amount of plastic waste are discharged into the environment, becoming a primary global environmental pollution problem. Under environmental conditions, plastic waste decomposes through multiple pathways, such as physical, chemical, and biological degradation, resulting in the slow release of tiny plastic debris.[2] Particularly, plastics debris smaller than 5 mm is one of the major sources of microplastics (MPs) that can be generated by plastic fragmentation through extensive weathering in the environment, and it also can be further fragmented into smaller particles, so-called nanoplastics (NPs) less than 1 μm in size. MPs/NPs can be generated either directly from industrial production as primary sources (e.g., cosmetics, personal care products, and synthetic textiles) or indirectly from the fragmentation of the large plastic debris as secondary sources through the natural weathering process (e.g., polyethylene (PE) or polypropylene (PP) fibers from the fishing net and water bottles).[3,4] In recent years, MPs/NPs have been found in several environmental media, including freshwater, oceans, soil, air, and even daily life products, such as cosmetics and tap water.[5] Thus, MPs/NPs have become a potential hazard to aquatic organisms and humans throughout the food supply. Moreover, humans could be exposed to plastic particles via inhalation, ingestion, or transdermal absorption, which may pose a substantial threat to human health, but the health effects have not been understood yet.[6]

Most plastic waste cannot degrade within a short period, and aging and weathering lead to polymer transformation and plastic degradation. The half-life of each plastic product in the ocean was estimated to range from 0.3 to 560 years in the presence of a natural degradation accelerant, including sunlight, heat, wave abrasion, biodegradation.[7] However, a long-term exposure to the natural degradation accelerant induces polymer transformation in plastics by cross-linking, chain scission, and oxidation via free radical formation and/or hydrolysis.[8] As a result, plastic properties can be changed in terms of mechanical strength, crystallinity, hydrophobicity, morphology, oxygen content, molecular weight, size, shape, and release of plastic additives that could be harmful to the environment and human health. The changed properties could impact the interaction of MPs/NPs with environmental contaminants, which may affect ingestion behavior and risk of plastic debris for small organisms.[9,10]

In the past decade, polymer degradation mechanisms have been extensively evaluated under different natural conditions. However, the biggest issue is the low degradation rate of plastics

in natural aging due to the low level of ultraviolet (UV), heat irradiation, and biological effects, resulting in the lack of understanding of long-term aging behaviour.[11] As an alternative to the natural aging, various laboratory technologies, such as UV irradiation, chemical oxidation, heat treatment, and microbial degradation, have been applied to accelerate the aging of plastics to clarify the degradation behavior from short to long periods in a simply operated and artificially-controlled manner.[10] For instance, the strong absorption of UV light (300–400 nm) with a high energy can alter the surface property of plastics in a short time. Song et al. reported that several years are required for the natural aging of PE, PP, and PS plastics to achieve the same aging degree with one-year exposure to UV-A lamp and mechanical abrasion, which causes cracking, discoloration, and significant fragmentation in the plastics. Chemical treatment is an alternative strategy to plastic degradation due to its powerful oxidizing potential.[12] Previous studies have shown that chemical oxidation, including Fenton, $K_2S_2O_8$, H_2O_2, and O_3, could effectively induce rapid aging for plastic recycling. Wang et al. reported that the mixtures of hydrogen peroxide (H_2O_2) and ferric oxide (Fe^{2+}) showed a relatively high oxidizing ability on PP and PE plastics at 25–160℃, called Fenton reactions, and generated MPs with different surface properties.[13] Although the oxidizing initiator or free radicals are artificially generated by the reaction, the natural aging caused by sunlight and heat also consists of free radical chain reactions.[14] The combination of chemical processing via catalysts and oxidants with high temperature is an alternative method of polymer degradation by accelerated aging of plastics via rapid oxidation in the polymer backbone. Moreover, the use of oxygen can significantly reduce heating in polymer degradation and can change the composition of the resulting products in a short time. In the case of the chemical recycling process, Zefirov et al. reported that the chemical PP degradation at a comparatively low temperature (150℃) in aqueous media with gaseous oxygen generated from H_2O_2 or the addition of CO_2 combined with H_2O_2 could enhance the efficiency of PP decomposition for MPs/NPs removal.[15] Recent review from Tian et al. reported that performing the oxidation reactions under high-pressure conditions can further accelerate radical diffusion and oxygen solubility within the polymer matrix, promoting deeper chain scission and oxidation. Consequently, the generated MPs and NPs exhibit distinct physicochemical characteristics—particle sizes, oxygen functionality, and surface roughness—compared with those formed at ambient pressure.[16] So far, several studies have summarized the degradation processes of PP, mainly considering the different aging processes and mechanisms.[13,17] However, systematic studies on PP degradation under combined H_2O_2 and high-pressure oxidation remain limited.

By investigating the oxidation of isotactic PP (iPP) pellets under high temperature, we previously showed the abundance of MPs/NPs in an oxygen-enriched medium with H_2O_2 under different thermal conditions.[18] In the present study, we investigated the thermal oxidation of iPP pellets at different hydrogen peroxide concentrations under high temperature and pressure and determined the properties of the resulting particles to estimate the optimal condition for removing MPs/NPs from plastic debris-containing media. The findings contribute to developing more efficient chemical upcycling and degradation strategies, minimizing persistent microplastic pollution, and supporting circular economy approaches for sustainable plastic waste management.

Materials and methods

Materials

Pellets of iPP (a number average molecular weight (M_n) of 97,000, Laboratory grade) were purchased from Sigma-Aldrich (St. Louis, Mo, USA). Hydrogen peroxide solution (H_2O_2, 30% (w/w)) and ethanol were purchased from FUJIFILM Wako Pure Chemical Corporation (Osaka, Japan). A vessel (50 mL) with a stainless-steel jacket and a polytetrafluoroethylene (PTFE) crucible body was purchased from Flon Industry Co., Ltd. (Tokyo, Japan). All chemicals were used without further purification. Milli-Q water was obtained using a Milli-Q Integral System (Millipore, a resistivity of 18.2 MΩ·cm at 25ºC).

Experiments

Thermal oxidation of iPP was conducted at 175℃ using a preheated oven as described previously.[18] Briefly, iPP pellets were immersed in H_2O_2/Milli-Q water mixtures of varying concentrations inside the reaction vessel, which was then sealed and rapidly transferred into the oven. After 24 h of the treatment, the vessel was cooled down to room temperature in the oven. The reaction product was centrifuged at 50,000 rpm for 15 min, and the solid fraction was washed with Milli-Q water to remove the non-solid one and by-product (residual acid) until the pH of the wash solution reached neutrality. The resulting solid product was dried in a desiccator for 1–2 days before weighting. After that, the dried solid product in each condition was resuspended in Milli-Q water and left overnight at room temperature for further analyses. As a control (g-MP), iPP pellets were ground into small pieces in Milli-Q water using a diamond polishing plate (REV, Roughness #320, READ Co., Ltd., Miyagi, Japan). The weight loss of iPP before and after the reaction was evaluated based on the following equation (Eq. 1)

$$Weight\ loss\ (\%) = (W_t - W_0)/W_0 \times 100 \quad (1)$$

where W_t is the weight of the solid product after reaction and W_0 is the initial weight of iPP pellets.

Surface morphology of the solid products was examined using scanning electron microscopy (SEM, SU-70, Hitachi High-Tech Corporation, Tokyo, Japan) at an electron-accelerating voltage of 3.0 kV. SEM images were analyzed by ImageJ software.

Zeta potential of the solid products in Milli-Q water was measured by electrophoretic light scattering (ELS) (SZ-100, HORIBA, Ltd., Tokyo, Japan).

Functional groups of the solid products were examined by Fourier Transform Infrared Spectroscopy (FTIR, IRPrestige21, SHIMADZU Corporation, Kyoto, Japan) with an

attenuated total reflection unit (ATR, DuraSamplIR, Smiths Detection, London, United Kingdom). The carbonyl index (CI) was calculated from the absorbance ratio between the carbonyl (C=O) peak at 1720 cm^{-1} and the methylene ($–CH_2$) peak at 1456 cm^{-1}.

Surface atomic ratio between carbon and oxygen for the solid products was evaluated from X-ray photoelectron spectroscopy (XPS, PHI5000 Versa Probe II, Kanazawa, Japan). Samples were mounted on an indium foil, and high-resolution C1s and O1s spectra were acquired with a pass energy of 58.7 eV, an energy step of 0.125 eV, and a detection angle of 45°. Based on the survey spectra, the O/C ratio was directly calculated from the integrated area of oxygen (532 eV) and carbon (284.9 eV) peaks. High-resolution XPS spectra were analyzed using MultiPak software.

Thermal properties and crystallinity of g-MP and the solid products were assessed using a differential scanning calorimeter (DSC 8230, RIGAKU Thermo plus EvoII, Tokyo, Japan) under a nitrogen (N_2) atmosphere. The initial heating rate was 20 ºC/min before 0℃, followed by 10℃/min between 0 and 200ºC. The degree of crystallinity is defined as follows (Eq. 2).

$$X_C = \Delta H_f(T_m) / \Delta H_f{}^0 T_m{}^0 \quad (2)$$

where X_c is the degree of crystallinity in weight; $\Delta H_f(T_m)$ is the enthalpy of fusion measured at melting temperature, T_m, and $\Delta H_f{}^0 T_m{}^0$ is the corresponding enthalpy of 100% crystalline polypropylene, which is usually 207 J/g.[19]

Polarized images of the solid products were visualized using a polarized optical microscope (Olympus BX53, Evident Techno Lab, Tokyo, Japan). A small amount of the solid products was cross sectioned using a Cryostat (Leica CM1950, Leica Biosystems, Tokyo, Japan) with a thickness of 1 µm. Polarized images were then recorded by converting unpolarized light to polarized light.

Pyrolysis–gas chromatography–mass spectrometry (Py-GC/MS) analyses were performed for iPP before and after the treatments. A small amount of each sample (0.1–1 mg) was placed initially into a Double-shot EGA/PY-3030D Pyrolyzer (Frontier Laboratories Ltd, Fukushima, Japan) using a Micro Jet Cryo-Trap (MJT-1030Ex) carrier gas selector. The pre-selected pyrolysis temperature was set at 700ºC, while the temperature of the GC oven was heated from 50 to 300ºC at 10ºC/min. The pyrolyzates were separated using a temperature-programmed capillary column of a Shimadzu GC-2030 gas chromatogram and analyzed by mass spectrometer MS-QP2020 NX (Shimadzu, Japan). The chromatogram and spectra retrieved by each experiment were subjected to further interpretation through Shimadzu GCMS post-run analysis software.

Results and discussion

In our previous work,[18] isotactic polypropylene (iPP) was oxidized using hydrogen peroxide under thermal treatment at 125, 150, and 175℃. Among these, 175℃ proved most effective for generating nanoscale particles, due to enhanced chain scission and surface oxidation. Thus, in the present study, the solid products were obtained at different concentrations of H_2O_2 ranging from 0 to 30% under pressure from approximately 2 to 9 bar at 175ºC for 24 h (Table S1) to evaluate the effect of H_2O_2 concentration on iPP degradation and their properties of the resulting solid product. Nevertheless, all obtained samples were significantly different visually, and a complete or significant decomposition of the iPP was observed [Fig. 1(a)]. When iPP pellets were treated with only aqueous media (0%), the sample became yellow-colored, but the sizes were still huge. This result could be due to the small amount of oxygen from air in the vessel. By increasing the concentration of H_2O_2 from 3 to 24%, dispersed black particles were obtained. The addition of H_2O_2 of more than 27% significantly changed the sample appearance, and the transparent and slightly colored liquid media was obtained. At a fixed temperature of 175ºC, varying the ratio of H_2O_2 to H_2O directly influences the internal pressure within the reaction vessel. This is primarily due to the thermal decomposition of H_2O_2 and H_2O, which releases molecular oxygen and contributes to pressure buildup. A higher amount of H_2O_2 leads to greater oxygen generation, thereby increasing the oxidative environment and internal pressure. This elevated pressure may accelerate oxidation rate and facilitate the formation of low-molecular-weight fragments, including MPs and NPs. However, it is important to emphasize that the observed effects are driven not by pressure alone, but by the presence of reactive oxygen species under pressurized conditions.[15]

Figure 1(b) shows the effect of hydrogen peroxide concentrations on weight loss of iPP in the treatment. Negligible weight loss occurred in the absence of H_2O_2, while a continuous increase in weight loss was observed up to the H_2O_2 concentration of 24%, reaching 100% weight loss at H_2O_2 concentrations exceeding 27%. The obtained weight loss under these conditions could be primarily attributed to the release of low-molecular-weight fragments and gaseous byproducts.[15] Notably, when the H_2O_2 exceeds 27%, complete weight loss is observed, indicating a critical threshold beyond where the polymer chains are completely cleaved or degradation products are fully volatized. Below this threshold, oxidation remains partial, with a significant portion of the material retained in the solid phase. Moreover, when the stability of H_2O_2 is reduced by heating and basic conditions, perhydroxyl anion (HOO^-) is formed, which then reacts with H_2O_2 to form the highly reactive hydroxyl radical (HO·), and this hydroxy radical could attack polymer chains to promote their degradation via random chain scission.[20] Besides, the addition of H_2O_2 as a degradation initiator could modulate the transport phenomenon in iPP, thereby altering the degradation kinetics by increasing the concentration of peroxide radicals (f) and the termination rate constant (K_t), both of which are key kinetic parameters in chemically initiated degradation of iPP.[21]

The charge at the shear plane on the surface of the solid products was measured as zeta potential (ζ), which is useful for

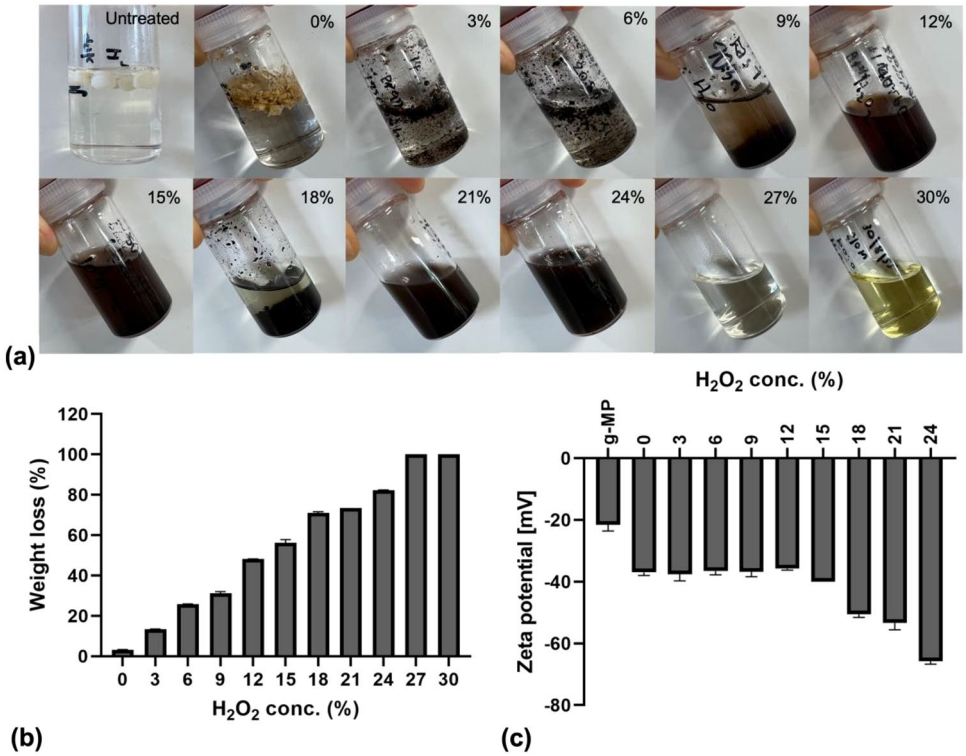

Figure 1. Appearance (a), weight loss (b) and zeta potential (c) of iPP treated at 175°C for 24 h with different H_2O_2 concentrations from 0–30 (%v/v). Values are mean ± SD.

understanding and predicting particle interactions in suspension [Fig. 1(c)]. The zeta potential for all treated samples decreased continuously from − 38 to − 65 mV with increasing H_2O_2 concentration, while untreated g-MP showed the zeta potential of − 23 mV. These results suggest that the increasing negative surface charge is likely due to the introduction of carbonyl and hydroxyl groups during the treatment.[22]

Figure 2(a) shows SEM images of the solid products obtained at different H_2O_2 concentrations. A control sample, untreated g-MP, showed relatively smooth surface, while rough surface appearance with small debris were seen on the pallet treated at H_2O_2 concentrations of 0 and 3%. On the other hand, when a larger amount of H_2O_2 was added into the reaction, the formation of small particles was observed. In addition, particle aggregation was seen at H_2O_2 concentrations ranging from 6 to 18%, probably due to contraction during cooling to room temperature. This behavior is attributed to the carbon chains attempting to fold together and reform ordered regions, despite potential changes in surface properties.[13] These phenomena might indicate the significant effect of H_2O_2 on the iPP degradation behavior. At a higher H_2O_2 concentration of 21%, some aggregated particles disappeared, which were consistent with the apparent weight loss. Moreover, nanoparticles were formed at H_2O_2 concentrations of 21% and 24%, possibly due to electrostatic repulsion between particles arising from surface charges introduced during the treatment.

Figure 2(b) shows the FTIR spectra of solid products before and after treatment at 175°C for 24 h under varying H_2O_2 concentrations. The characteristic peaks at 2915 and 2847 cm^{-1}, corresponding to the asymmetric and symmetric stretching vibrations of CH_2 groups, respectively, disappeared in the product obtained at 24% H_2O_2. In addition, the peaks at 1473 and 1463 cm^{-1}, attributed to the bending deformation of CH_2 and CH_3 groups, were significantly reduced at H_2O_2 concentrations of 21% and 24%, respectively. Conversely, a broad absorption band in the range of 3300–3600 cm^{-1}, corresponding to the O–H stretching vibration, appeared at higher H_2O_2 concentrations, along with peaks at 1714 and 1587 cm^{-1}, which are attributed to the C=O stretching of carboxylic acids or ketones and C–OH bending modes, respectively. Additional broad peaks at 1199 and 1067 cm^{-1} also emerged after treatment, suggesting the presence of functional groups such as C–O–C (typical of ethers or lipids) and C–O (associated with alcohols).[23] FTIR spectra show progressive growth of C=O, O–H, and C–O bands alongside attenuation of iPP backbone/crystallinity bands, evidencing hydroperoxide formation and subsequent β-scission of alkoxy radicals to yield carbonyl-terminated oligomers and secondary oxidation products. These findings suggest that iPP was oxidized through the treatment.

Furthermore, the absorption bands in the regions associated with C=O groups were observed in each sample, although the maximum intensity of the carbonyl peaks was different. Specifically, the peak intensity of C=O appeared and dramatically increased during the treatment. On the other hand, the peaks of CH_2 and CH_3 groups broadened with an increase in

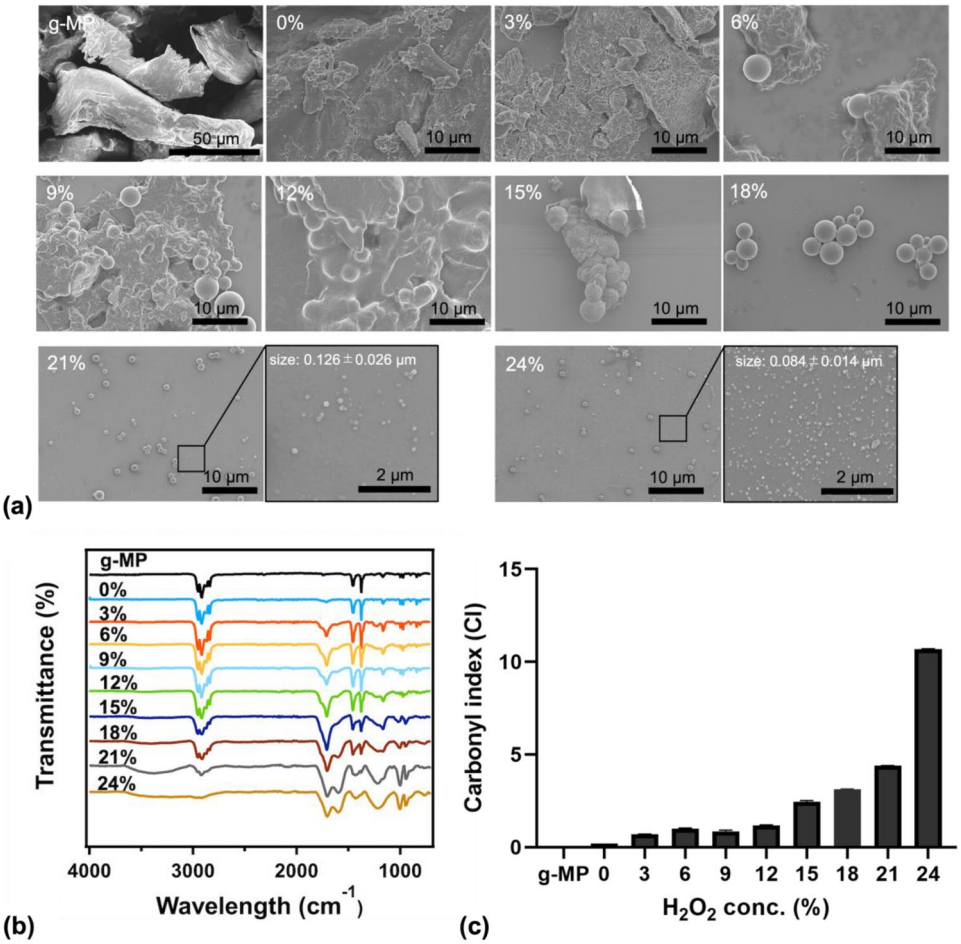

Figure 2. SEM images (a), FTIR spectra (b) and carbonyl index (c) of g-MP and solid products obtained after treatment at 175℃ for 24 h with varying H_2O_2 concentrations from 0 to 24 (%v/v). The scale bars are 50, 10 and 2 μm, respectively. Values are mean ± SD.

the H_2O_2 concentration, demonstrating chain scission. Besides, the degree of oxidation indicating the aging degree of plastics was further evaluated using the carbonyl index (CI) calculated as the ratio of the absorbance of the C=O peak (1716 cm^{-1}) to that of the methylene (CH_2) scissoring peak (1473 cm^{-1}), [24] as shown in Fig. 2(c). The value of CI increased continuously with simulated H_2O_2 concentration up to approximately 12, in agreement with the FTIR results. The sharp rise in carbonyl index at ≥ 3–18% H_2O_2 indicates entry into an autocatalytic regime of scission/oxidation, and at 21–24% the polymer is extensively fragmented to low-molecular-weight oxidized species, explaining the loss of recoverable solid. The untreated control sample (g-MP) had little effect on the change in the surface morphology and functional groups of iPP due to natural light irradiation. These results suggest that the oxidation degree of iPP varied significantly with H_2O_2 concentration due to its high susceptibility to oxygen attack. The presence of hydroxyl radicals, known to accelerate MP/NP degradation, likely played a critical role. Moreover, the formation of carbonyl groups is a direct consequence of oxidative chain scission during polypropylene degradation. As the polymer backbone undergoes β-scission in the presence of oxygen, carbonyl-containing end groups are generated at the cleavage sites. Therefore, a high level of carbonyl formation as evidenced by FTIR analysis can be interpreted as a reliable indicator of extensive chain cleavage and molecular weight reduction.[25]

The oxidation process could trigger a chemical alteration of the surface of plastics with polymer chain scission, creating a double bond and/or bonds to more than one atom. The element-binding energy of C1s and O1s in XPS analyses for the iPP surface before and after thermal oxidation is shown in Fig. S1. The results showed that the C1s spectrum contained two peaks, C–C and C=O, suggesting that there are regions of thermally decomposed and natural light degradation in the untreated g-MP. After being treated with H_2O_2, the C1s spectrum corresponding to C–C, C–O–C/C–OH, C=O, and O=C–O were detected at 284.5, 286.3, 287.3, and 289.0 eV, respectively. The C–C peak area decreased, while the C–O–C/C–OH, C=O, and O=C–O peak area increased along with the concentration of H_2O_2, implying chain scission and chain end oxidation in iPP

after thermal oxidation. The C1s and O1s content and the relative degree of oxidation (O/C ratio) are shown in Fig. 3(a). The O/C ratio of oxidized isotactic polypropylene (OPP) dramatically increased from below 0.1 to 0.35 depending on the concentration of H_2O_2. Similar to the carbonyl index, the oxygen content of OPP increased to 30% with the H_2O_2 concentration of 24%, while untreated g-MP showed a carbon and oxygen content of 90% and 10%, respectively. Notably, the surface of OPPs became hydrophilic due to the surface oxidation with the increasing O/C ratio.[26]

The thermal characteristics of OPPs were evaluated by DSC measurement in a wide range of temperatures [Figs. 3(b and c)], and the crystallinity of all OPPs is calculated from Eq. 2. The DSC curves of OPPs treated with different H_2O_2 concentrations showed that the degree of crystallinity decreased until it disappeared (OPP6), as shown in Table I. When the sample was cooled down slowly from the hydrothermal condition (175ºC), the carbon chains in the amorphous phase usually facilitate the crystallization process. However, at an H_2O_2 concentration below 6%, the level of generated carbonyl groups was not high enough to disrupt the crystallization process. On the other hand, the crystallinity disappeared after being treated with a high concentration of H_2O_2 (more than 9%). This phenomenon with the oxidation somehow impaired the crystallization of iPP, which is related to the increased irregularity of carbon chains as well as the increased electrostatic repulsion caused by the increment of carbonyl groups. Thus, the decreased crystallinity suggests that the newly formed carbonyl groups could inhibit the recrystallization due to the steric hindrance by increasing the interplanar spacing and ultimately restraining the stretched carbon chains to form compact structures.[27] This behavior is consistent with the findings reported by Wang et al. [28] Moreover, the loss of crystallinity reflects polymer chain scission because the cleavage of polymer backbones disrupts the regular chain alignment required for crystalline domain formation. As oxidation

Table I. Calculated results from DSC analysis.

Samples	g-MP	OPP0	OPP3	OPP6	OPP9
Enthalpy (J g^{-1})	71.8	65.9	11.2	6.2	N/A
Crystallinity (%)	34.7	31.8	5.3	3.0	N/A

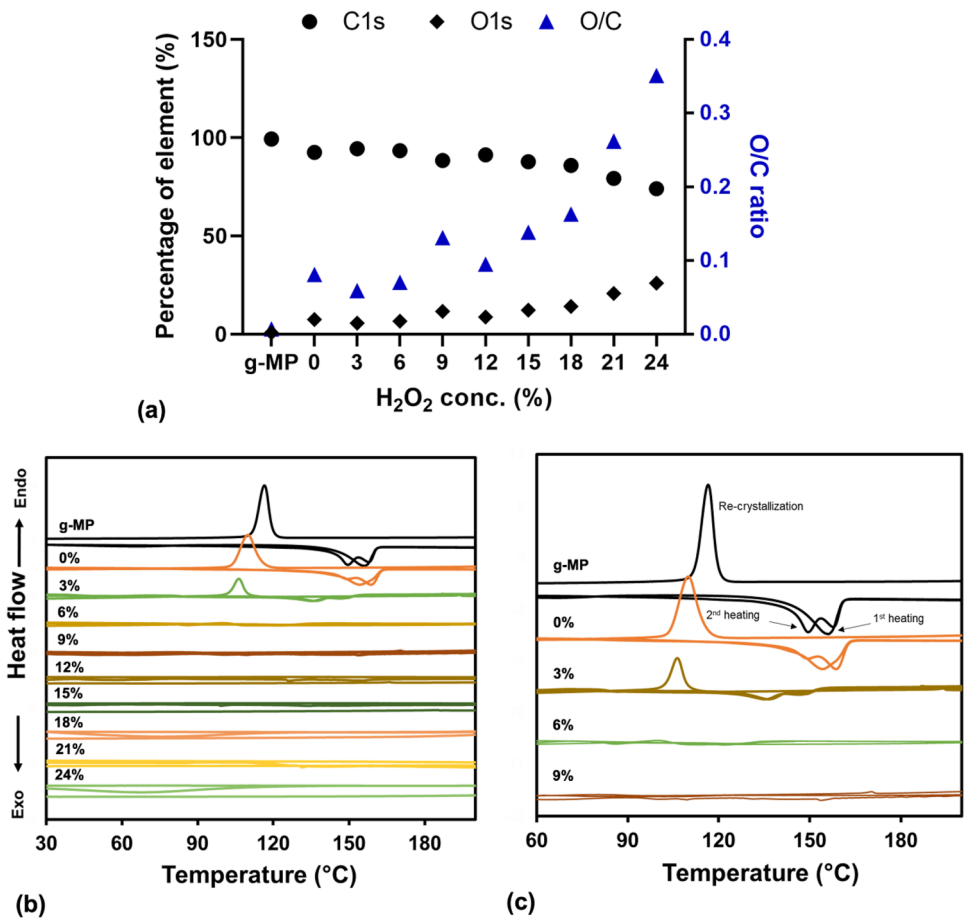

Figure 3. Atomic concentration (%) and O/C ratio (a), DSC heating–cooling curves (b) and g-MP and OPP0, OPP3, OPP6, and OPP9 (c) of untreated g-MP and OPPs obtained at different H_2O_2 concentration from 0 to 24 (%v/v).

proceeds, long isotactic polypropylene chains are broken into shorter segments, reducing molecular weight and hindering their ability to fold and pack into ordered lamellae. Consequently, the amorphous fraction increases while the crystalline regions diminish, indicating progressive chain degradation.

Consistent with the DSC crystallinity data, the polarized optical microscopic observation showed that the spherulite size of OPPs decreases quickly with increasing H_2O_2 concentrations, while the large spherulites exist in untreated g-MP (Fig. S2). These results demonstrate that the different oxidation levels can effectively modulate the formation of spherulites.

Polymer decomposition has been widely evaluated by pyrolysis coupled with GC/MS as it provides the chemical structure of macromolecules after analyzing its evolved thermal degradation products. The characteristic thermal degradation compounds of untreated g-MP and treated samples with the characteristic fragment ions and their retention times are recorded in Fig. S3 and Table 2S-11S. Concerning the untreated g-MP pyrolytic profile, the most abundant pyrolysis products are 2,4 dimethyl-1-heptane along with other propylene oligomers in agreement with well-known pyrolytic products of iPP-related random C–C scissions followed by intramolecular hydrogen transfer yielding alkanes and 1-alkenes. OPPs with a higher H_2O_2 concentration showed an increase in oxidation products or degradation markers, such as carbon dioxide, acetone, and acetic acid, but a decrease in 2,4 dimethyl-1-heptane, indicating chain scission reactions and formation of carbonyls characteristic of the oxidation reaction.[29] H_2O_2 at a higher concentration could stimulate the initial cleavage of larger hydrocarbon molecules into smaller ones as well as the formation of other molecules that are vulnerable at the site of thermal decomposition, thus potentially leading to the decomposition of shorter hydrocarbon chains.[30,31] The qualitative assessment of pyrolysis products coming from untreated g-MP and OPPs agreed with those previously reported in studies looking at the significant reduction of molecular weight of OPPs by a thermal oxidation with different heating temperatures compared to virgin iPP (Fig. S4)[18].

Together, FTIR, DSC, and Py-GC/MS collectively indicate radical oxidation–driven chain scission in iPP with increasing H_2O_2. FTIR shows a monotonic rise of C=O and O–H/C–O bands, evidencing the buildup of oxidized functionalities. Consistently, DSC reveals a progressive reduction and broadening of the melting endotherm, a slight depression of T_m, and loss of recrystallization, reflecting reduced lamellar thickness and crystallinity caused by scission and defect incorporation. Py-GC/MS further corroborates this chemistry: the relative abundance of early-eluting oxygenated volatiles increases while hydrocarbon oligomer peaks diminish, indicating that the oxidized, lower-molecular-weight matrix preferentially decomposes into oxygen-bearing fragments. The concurrent evolution across all three techniques supports a mechanism of hydroperoxide formation, alkoxy-radical β-scission, and secondary oxidation that accelerates at $\geq 6\%$ H_2O_2.

Conclusion

In the aqueous environment, MPs/NPs can cause potential physical injuries to microorganisms and act as a vector of detrimental pollutants by food chains. Even though current technologies can eliminate and detect MPs but not NPs, a large amount of MPs/NPs still remain in the aqueous environment. MPs/NPs remediation strategies in an aqueous environment are needed to evaluate the effective plastic degradation. This study demonstrated that isotactic polypropylene (iPP) can be efficiently degraded through high-pressure hydrothermal oxidation using hydrogen peroxide. Increasing H_2O_2 concentration from 3 to 24% progressively enhanced chain scission, surface oxidation, and crystallinity loss, while complete mineralization occurred at concentrations above 27%. Weight-loss analysis confirmed 100% degradation above this threshold, consistent with the disappearance of solid residues and the formation of a clear solution. SEM, FT-IR, and XPS analyses revealed the generation of carbonyl and hydroxyl groups, an O/C ratio increase up to 0.35, and a carbonyl index of ≈ 12, indicating extensive oxidation. DSC results showed a reduction in crystallinity from 34.7% (g-MP) to below 3%. These results confirm that high-pressure H_2O_2 oxidation effectively converts iPP into oxidized fragments and gaseous products within 24 h, offering a practical and scalable approach for integrating hydrothermal oxidation into tertiary wastewater treatment to remove residual MPs/NPs.

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Author contributions

S. H., N.W., A.K., T.T., T.K., and M.Y. conceived and designed the study. S.H., N.W., M. K., T.T., T.K., and M.Y. contributed to the data acquisition, data analysis, data interpretation, and manuscript writing. S.F. contributed to data interpretation. All authors contributed to manuscript editing. M.Y. supervised the study and acquired funding. All authors critically revised and approved the final manuscript for publication.

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Data availability

The authors declare that the data supporting the findings of this study are available within the paper and its Supplementary Information files. Should any raw data files be needed in another format they are available from the corresponding author upon reasonable request.

Declarations

Competing interests

The authors declare no completing financial interest.

Supplementary Information

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